Literature Search Log

Biological Efficacy of New Insecticides and Side Effects on Beneficial Entomo- and Acarofauna in Greenhouses

Compiled for Nick's PhD dissertation research (Agricultural University of Plovdiv, supervisor Assoc. Prof. Dr. Dima Markova).

Search date: 2026-07-29. Sources: Google Scholar, PubMed, Semantic Scholar, ResearchGate, Wiley, Springer, MDPI, ScienceDirect, Frontiers, Hrcak, Bulgarian Journal of Agricultural Science.

HIGH PRIORITY — Published within the last 6 months (Jan–Jul 2026)

Sulfoxaflor Modulates Diet-Dependent Effects on Bumble Bee Development but Shows No Detectable Effects on Adult Respiration Rate and Patterns

HIGH

Authors:	Jürison, M., Teras, P., Pent, K., et al.
Year:	2026
Journal:	Scientific Reports
DOI/PMID:	PMID 42426207
Free full text:	Scientific Reports is fully open access — https://www.nature.com/srep/ (search PMID 42426207 for direct link)
Relevance:	Sulfoxaflor side effects on beneficial pollinators (bumble bees used in greenhouse tomato pollination) — sublethal/developmental toxicity
APA Citation:	Jürison, M., Teras, P., Pent, K., et al. (2026). Sulfoxaflor modulates diet-dependent effects on bumble bee development but shows no detectable effects on adult respiration rate and patterns. *Scientific Reports*.

Sulfoxaflor Reduces Food Intake and Learning Efficiency of Solitary Osmia bicornis Bees

HIGH

Authors:	Schwarz, J. M., Arnet, N. L., Knauer, A. C., Michez, D., & Albrecht, M.
Year:	2026
Journal:	Scientific Reports
DOI/PMID:	PMID 42236841
Free full text:	Open access — Scientific Reports (Nature)
Relevance:	Sulfoxaflor sublethal effects on beneficial pollinating insects; relevant to broader "side effects on beneficial entomofauna" theme
APA Citation:	Schwarz, J. M., Arnet, N. L., Knauer, A. C., Michez, D., & Albrecht, M. (2026). Sulfoxaflor reduces food intake and learning efficiency of solitary Osmia bicornis bees. *Scientific Reports*.

Sustainable Management of Whitefly (Bemisia tabaci) in Vegetables Using Entomopathogenic Fungi, Isaria farinosa: Characterization, Efficacy, Insecticide Compatibility with Molecular Docking Insights

HIGH

Authors:	Manjunath, M., Kodandaram, M. H., Divekar, P. A., et al.
Year:	2026
Journal:	World Journal of Microbiology and Biotechnology
DOI/PMID:	PMID 42334685
Free full text:	Abstract only confirmed — check publisher (Springer) for open access status
Relevance:	Bemisia tabaci management; compatibility of cyantraniliprole and sulfoxaflor with entomopathogenic fungi — directly relevant to integrating chemical and biological control in vegetable production

APA Citation:

Manjunath, M., Kodandaram, M. H., Divekar, P. A., et al. (2026). Sustainable management of whitefly (Bemisia tabaci) in vegetables using entomopathogenic fungi, Isaria farinosa: Characterization, efficacy, insecticide compatibility with molecular docking insights. *World Journal of Microbiology and Biotechnology*.

Efficacy of Insecticide Application Against Aphis gossypii and Its Influence on the Predatory Capacity of Hippodamia variegata

HIGH

Authors:	Li, P., et al. (Tarim University research group)
Year:	2026
Journal:	Agronomy, 16(2), 228
DOI:	10.3390/agronomy16020228
Free full text:	YES — MDPI open access journal: https://www.mdpi.com/2073-4395/16/2/228
Relevance:	Direct comparison of afidopyropen and flonicamid efficacy against Aphis gossypii plus selectivity/side effects on the predatory ladybird Hippodamia variegata (selectivity toxicity ratio: afidopyropen 5.05, flonicamid 4.73; field control 96–97%)
APA Citation:	Li, P., et al. (2026). Efficacy of insecticide application against Aphis gossypii and its influence on the predatory capacity of Hippodamia variegata. *Agronomy, 16*(2), 228. https://doi.org/10.3390/agronomy16020228

Pesticide-Induced Lethal and Sublethal Effects in Predatory Mites: A Review

HIGH

Authors:	Sharma, S., Sharma, P., Sanjta, S., et al.
Year:	2026
Journal:	Phytoparasitica, 54, 42
DOI:	10.1007/s12600-026-01375-x
Free full text:	Abstract only confirmed via Springer link — check for open access
Relevance:	Comprehensive review of pesticide lethal/sublethal effects on Phytoseiidae and other predatory mites (beneficial acarofauna) — excellent framing/background reference for Nick's acarofauna chapter
APA Citation:	Sharma, S., Sharma, P., Sanjta, S., et al. (2026). Pesticide-induced lethal and sublethal effects in predatory mites: A review. *Phytoparasitica, 54*, 42. https://doi.org/10.1007/s12600-026-01375-x

HIGH / MEDIUM PRIORITY — Published 2024–2025 (within 5-year window)

Side Effects of Semi-Synthetic Insecticide Spinetoram on the Whitefly Parasitoid Encarsia formosa

HIGH

Authors:	Drobnjaković, T., Prijović, M., Dervišević, M., Brkić, D., Ricupero, M., & Marčić, D.
Year:	2025
Journal:	Pest Management Science, 81, 490–497
DOI:	10.1002/ps.8450
Free full text:	Abstract only — paywall (Wiley). Preprint/author copy may exist on ResearchGate: https://www.researchgate.net/publication/384429893
Relevance:	Direct core topic — lethal/sublethal side effects of a newer insecticide on Encarsia formosa, the key greenhouse whitefly parasitoid central to Nick's dissertation
APA Citation:	Drobnjaković, T., Prijović, M., Dervišević, M., Brkić, D., Ricupero, M., & Marčić, D. (2025). Side effects of semi-synthetic insecticide spinetoram on the whitefly parasitoid Encarsia formosa. *Pest Management Science, 81*, 490–497. https://doi.org/10.1002/ps.8450

Control of Two Sucking Insect Pests, a Whitefly (Bemisia tabaci) and a Thrips (Frankliniella occidentalis), Infesting Hot Peppers by Spraying LDH-Formulated dsRNA

MEDIUM-HIGH

Authors:	Khan, R. A. A., et al.
Year:	2025 (published August 2025)
Journal:	Pest Management Science
DOI:	10.1002/ps.70125
Free full text:	YES — PMC12618903: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12618903/
Relevance:	Novel RNAi-based alternative/complement to chemical insecticides against two of Nick's core target pests (B. tabaci, F. occidentalis) in greenhouse peppers — good for discussion/future-directions section
APA Citation:	Khan, R. A. A., et al. (2025). Control of two sucking insect pests, a whitefly (Bemisia tabaci) and a thrips (Frankliniella occidentalis), infesting hot peppers by spraying LDH-formulated dsRNA. *Pest Management Science*. https://doi.org/10.1002/ps.70125

Pollen Provisioning Attenuates Pesticide Side-Effects on a Phytoseiid Predator

MEDIUM-HIGH

Authors:	Samaras, K., Pappas, M. L., Pozzebon, A., & Broufas, G. D.
Year:	2024
Journal:	Pest Management Science, 80(6), 2619–2625
DOI:	10.1002/ps.7969
Free full text:	Abstract only — paywall (Wiley)
Relevance:	Flonicamid side effects on predatory mite Amblydromalus limonicus (Phytoseiidae)

	and how supplemental pollen feeding mitigates sublethal impacts — directly relevant to IPM/acarofauna conservation chapter
APA Citation:	Samaras, K., Pappas, M. L., Pozzebon, A., & Broufas, G. D. (2024). Pollen provisioning attenuates pesticide side-effects on a phytoseiid predator. *Pest Management Science, 80*(6), 2619–2625. https://doi.org/10.1002/ps.7969

Toxicity and Risk Assessment of Pesticides on the Predatory Mite, Amblyseius swirskii Athias-Henriot (Acari: Phytoseiidae) Under Laboratory Conditions

MEDIUM-HIGH

Authors:	Döker, İ., Yavaş, H., Shirvani, Z., Karaca, M. M., Karut, K., Marčić, D., & Kazak, C.
Year:	2024
Journal:	Phytoparasitica, 52, 97
DOI:	10.1007/s12600-024-01217-8
Free full text:	Abstract only — check Springer for open access
Relevance:	IOBC-style risk assessment of several "reduced-risk" pesticides on the key beneficial predatory mite Amblyseius swirskii; spinosad/spinetoram highly toxic vs. fungicides less so — directly usable methodology template for Nick's own acarofauna bioassays
APA Citation:	Döker, İ., Yavaş, H., Shirvani, Z., Karaca, M. M., Karut, K., Marčić, D., & Kazak, C. (2024). Toxicity and risk assessment of pesticides on the predatory mite, Amblyseius swirskii Athias-Henriot (Acari: Phytoseiidae) under laboratory conditions. *Phytoparasitica, 52*, 97. https://doi.org/10.1007/s12600-024-01217-8

Proactive Resistance Management Studies Highlight the Role of Cytochrome P450 Genes in the Resistance of Tuta absoluta Against Tetraniliprole

HIGH

Authors:	Ullah, F., Ullah, Z., Gul, H., Li, X., Pan, Y., Zhang, H., Zhang, Z., Huang, J., Roditakis, E., Guedes, R. N. C., Desneux, N., & Lu, Y.
Year:	2025 (published 28 May 2025)
Journal:	International Journal of Molecular Sciences, 26(11), 5180
DOI:	10.3390/ijms26115180
Free full text:	YES — MDPI open access: https://www.mdpi.com/1422-0067/26/11/5180 ; also PMC12155287
Relevance:	Diamide-class insecticide (structurally/mechanistically related to chlorantraniliprole/cyantraniliprole) resistance mechanisms in Tuta absoluta — directly relevant to resistance-management discussion for Nick's target pest
APA Citation:	Ullah, F., Ullah, Z., Gul, H., Li, X., Pan, Y., Zhang, H., Zhang, Z., Huang, J., Roditakis, E., Guedes, R. N. C., Desneux, N., & Lu, Y. (2025). Proactive resistance management studies highlight the role of cytochrome P450 genes in the resistance of Tuta absoluta against tetraniliprole. *International Journal of Molecular Sciences, 26*(11), 5180. https://doi.org/10.3390/ijms26115180

Nanocarrier-Mediated RNAi of CYP9A306 (CYP9E2) and CYB5R Enhances Susceptibility of Invasive Tomato Pest, Tuta absoluta to Cyantraniliprole

HIGH

Authors:	Ullah, F., Guru-Pirasanna-Pandi, G., Gul, H., Panda, R. M., Murtaza, G., Zhang, Z., Huang, J., Li, X., Desneux, N., & Lu, Y.
Year:	2025 (published 28 April 2025; corrigendum issued)
Journal:	Frontiers in Plant Science, 16
DOI:	10.3389/fpls.2025.1573634
Free full text:	YES — Frontiers is fully open access: https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2025.1573634/full ; PMC12066504
Relevance:	Cyantraniliprole resistance mechanism and RNAi-enhanced susceptibility in Tuta absoluta — directly on-topic for insecticide efficacy/resistance chapter
APA Citation:	Ullah, F., Guru-Pirasanna-Pandi, G., Gul, H., Panda, R. M., Murtaza, G., Zhang, Z., Huang, J., Li, X., Desneux, N., & Lu, Y. (2025). Nanocarrier-mediated RNAi of CYP9A306 and CYB5R enhances susceptibility of invasive tomato pest, Tuta absoluta to cyantraniliprole. *Frontiers in Plant Science, 16*, 1573634. https://doi.org/10.3389/fpls.2025.1573634

Susceptibility of Different Egyptian Populations of Tuta absoluta (Meyrick) to Selected Insecticides and Associated Detoxification Enzyme Activities

MEDIUM

Authors:	[Egyptian research group — full author list to be confirmed on publisher page]
Year:	2025
Journal:	Insects, 17(5), 493
DOI:	10.3390/insects17050493
Free full text:	YES — MDPI open access
Relevance:	Baseline insecticide susceptibility monitoring in Tuta absoluta field populations, including diamide-class compounds
APA Citation:	[Authors]. (2025). Susceptibility of different Egyptian populations of Tuta absoluta (Meyrick) to selected insecticides and associated detoxification enzyme activities. *Insects, 17*(5), 493. https://doi.org/10.3390/insects17050493

Note: Note: full author byline not yet independently confirmed — verify on MDPI page before citing in the dissertation.

MEDIUM PRIORITY — Solid background/comparative papers (2020–2024)

Life Parameters and Physiological Reactions of Cotton Aphids Aphis gossypii (Hemiptera: Aphididae) to Sublethal Concentrations of Afidopyropen

MEDIUM

Authors:	[research group, Tarim University-affiliated]
Year:	2024
Journal:	Agronomy, 14(2), 258
DOI:	10.3390/agronomy14020258
Free full text:	YES — MDPI open access
Relevance:	Sublethal physiological effects of afidopyropen on Aphis gossypii, one of Nick's core target pests
APA Citation:	[Authors]. (2024). Life parameters and physiological reactions of cotton aphids Aphis gossypii (Hemiptera: Aphididae) to sublethal concentrations of afidopyropen. *Agronomy, 14*(2), 258. https://doi.org/10.3390/agronomy14020258

Učinak spiropidiona na lisne uši i predatorsku grinju Amblyseius swirskii u uzgoju paprike [Effect of spiropidion on aphids and the predatory mite Amblyseius swirskii in pepper cultivation]

MEDIUM-HIGH

Authors:	Bažok, R., Čačija, M., Kadoić Balaško, M., Lemić, D., Damjanović, L., Skendžić, S., Drmić, Z., & Jelovčan, S.
Year:	2021
Journal:	Glasilo biljne zaštite (Croatian Plant Protection Bulletin), 21(5), 523–532
DOI:	Not assigned (Croatian national journal)
Free full text:	YES — Hrcak repository: https://hrcak.srce.hr/263592 (open access PDF, CC BY-NC)
Relevance:	Direct match to Nick's core interest — efficacy of spiropidion (one of his 7 target insecticides) against aphids AND side effects on Amblyseius swirskii (key beneficial acarofauna) in protected/greenhouse pepper cultivation
APA Citation:	Bažok, R., Čačija, M., Kadoić Balaško, M., Lemić, D., Damjanović, L., Skendžić, S., Drmić, Z., & Jelovčan, S. (2021). Učinak spiropidiona na lisne uši i predatorsku grinju Amblyseius swirskii u uzgoju paprike [Effect of spiropidion on aphids and the predatory mite Amblyseius swirskii in pepper cultivation]. *Glasilo biljne zaštite, 21*(5), 523–532.

Identification and Functional Validation of Glutathione S-Transferase Genes Involved in Detoxification of Sulfoxaflor, Afidopyropen and Lambda-Cyhalothrin in Aphis glycines

MEDIUM

Authors:	Guan, N., Dang, F., Qiu, S., & Xia, J.
Year:	2026
Journal:	Pest Management Science

DOI/PMID:

	PMID 42438111
Free full text:	Abstract only — paywall likely (Wiley)
Relevance:	Detoxification mechanisms for sulfoxaflor and afidopyropen (both core target insecticides), though studied in soybean aphid rather than a greenhouse pest — useful mechanistic background
APA Citation:	Guan, N., Dang, F., Qiu, S., & Xia, J. (2026). Identification and functional validation of glutathione S-transferase genes involved in detoxification of sulfoxaflor, afidopyropen and lambda-cyhalothrin in Aphis glycines. *Pest Management Science*.

The Effects of Natural Insecticides on the Green Peach Aphid Myzus persicae (Sulzer) and Its Natural Enemies Propylea quatuordecimpunctata (L.) and Aphidius colemani Viereck

MEDIUM

Authors:	[confirm full author list on publisher page]
Year:	2024 (published 22 July 2024)
Journal:	Insects, 15(7), 556
DOI:	10.3390/insects15070556
Free full text:	YES — MDPI open access / PMC11277335: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11277335/
Relevance:	Selectivity of botanical/soft insecticides toward beneficial entomofauna (ladybird + parasitoid) attacking aphids — useful comparative/control benchmark against the synthetic target insecticides
APA Citation:	[Authors]. (2024). The effects of natural insecticides on the green peach aphid Myzus persicae (Sulzer) and its natural enemies Propylea quatuordecimpunctata (L.) and Aphidius colemani Viereck. *Insects, 15*(7), 556. https://doi.org/10.3390/insects15070556

Insecticidal Activity of Some Botanical Products Against Green Peach Aphid (Myzus persicae Sulzer) and Thrips (Thrips tabaci Lindeman, Frankliniella occidentalis Perg.) in Greenhouse Pepper Production

HIGH (PENDING VERIFICATION)

Authors:	Yankova, V., Markova, D., & Todorova, V.
Year:	2025 (indexed publication date 24 June 2025)
Journal:	Bulgarian Journal of Agricultural Science
DOI:	Not confirmed — direct primary-source page could not be reached (site access blocked during this search session)
Free full text:	Likely YES via agrojournal.org, but the direct PDF link could not be verified this session — RECOMMEND Nick/Dr. Markova confirm directly via the journal or institutional access, since this is from his own supervisor's group
Relevance:	Directly from Nick's supervisor's research group; covers two of his core target pest species (Thrips tabaci, Frankliniella occidentalis) in greenhouse pepper — very likely citable as institutional prior work, but VERIFY before formal citation given the DOI/PDF could not be independently confirmed today
APA Citation:	Yankova, V., Markova, D., & Todorova, V. (2025). Insecticidal activity of some botanical products against green peach aphid (Myzus persicae Sulzer) and thrips (Thrips tabaci Lindeman, Frankliniella occidentalis Perg.) in greenhouse pepper

production. *Bulgarian Journal of Agricultural Science*. [volume/pages/DOI to be confirmed]

LOW PRIORITY / Background Only

Selectivity of Few Chemical Insecticides and Entomopathogenic Fungi Cordyceps fumosorosea on the Parasitoid Encarsia guadeloupae and on the Predator Apertochrysa astur in Coconut

LOW

Authors: [Indian NBAIR research group, incl. K. Selvaraj]

Year: 2026

Journal: BioControl

DOI: 10.1007/s10526-026-10385-x

Free full text: Abstract only — paywall likely

Relevance: Tests flupyradifurone (one of Nick's target insecticides) and spirotetramat for selectivity toward a whitefly parasitoid and a chrysopid predator — same conceptual framework as Nick's dissertation, but conducted in coconut, not greenhouse vegetables

APA Citation: [Authors]. (2026). Selectivity of few chemical insecticides and entomopathogenic fungi Cordyceps fumosorosea on the parasitoid Encarsia guadeloupae and on the predator Apertochrysa astur in coconut. *BioControl*. https://doi.org/10.1007/s10526-026-10385-x

Fate and Effects of Flupyradifurone and Sulfoxaflor on Non-Target Soil-Dwelling Invertebrates

LOW

Authors: [confirm on publisher page]

Year: 2026

Journal: Environmental Toxicology and Chemistry, 45(6), 1494

DOI: 10.1093/etojnl/vgag053

Free full text: Abstract only — paywall (Oxford Academic)

Relevance: Non-target effects of two of Nick's core insecticides, though on soil invertebrates (Folsomia candida) rather than entomo-/acarofauna proper — tangential background

APA Citation: [Authors]. (2026). Fate and effects of flupyradifurone and sulfoxaflor on non-target soil-dwelling invertebrates. *Environmental Toxicology and Chemistry, 45*(6), 1494. https://doi.org/10.1093/etojnl/vgag053

Notes on Search Process

- Full author lists for several very recent (2026) papers could not be completely verified past first 2–4 authors due to search-engine summarization; before final citation in the dissertation, Nick should pull the full reference (all authors, volume/issue/page numbers) directly from PubMed/publisher pages using the DOIs/PMIDs listed here.
- The Yankova/Markova/Todorova (2025) paper is flagged HIGH priority given its direct relevance and institutional connection, but its DOI and exact page numbers could not be independently confirmed this session (agrojournal.org blocked automated access with a bot-verification page). Recommend direct retrieval via institutional login or by asking Dr. Markova for the reprint.
- No papers were fabricated; several near-relevant candidates on Sulfoxaflor + Encarsia formosa and Flupyradifurone + green lacewings (2023 Nature Scientific Reports and Chrysoperla carnea studies) were found but are outside the 2024–2026 window and were left out of this log to keep focus on current literature — available on request for the background/lit-review chapter.